



Optimizing nutrient requirement in space and time for increased crop yield and income

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Received: 23 January 2025 / Accepted: 28 July 2025 / Published online: 22 August 2025
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Abstract The rapidly increasing population growth in sub-Saharan Africa (SSA), disruptions in the fertilizer supply chain due to the COVID-19 pandemic, and recent geopolitical tensions have highlighted the need for immediate actions toward optimizing fertilizer nutrients use. Fertilizer recommendations in SSA are either pan-territorial, obsolete, having been developed in the 1970s or 1980s, or completely non-existent for major crops, including maize. The International Fertilizer Development Center (IFDC), with the mission to improve soil health through better nutrient management, has developed and used soil and crop simulation models over the years to capture complex soil–plant–atmosphere interactions and their effect on crop response to nutrient application. IFDC developed the Geographic Support System for Agrotechnology Transfer (GSSAT2) software, an enhanced version of the Decision Support System

for Agrotechnology Transfer, that integrates features for autonomous data collection, simulation parallelism, and intuitive geographic information system data visualization, in combination with economic analysis, to derive geospatial fertilizer recommendations. The GSSAT2 approach has been adopted by the Sustainable Opportunities for Improving Livelihoods with Soils Consortium Space to Place initiative to develop more effective and localized soil fertility recommendations for more profitable, productive, and environmentally sustainable fertilizer use by smallholder farmers. Application of GSSAT2 using legacy data from Zambia on weather, soil physical and chemical properties, crop genetics and management, and input and output prices in maize revealed high spatial and temporal variability of nutrient requirements and yields, justifying the need for fertilizer recommendations targeted to farmers' conditions. November and December appeared to be the best months for planting maize, with rainfed potential yields ranging from 5.5 to 6.8 Mg ha⁻¹. Earlier planting could work in Western Zambia, while later planting in January and February could occur in selected parts of Eastern Zambia, but with high risks of crop failure due to drought. In addition to assessing tailored nutrient management capabilities using GSSAT2, this paper also discusses further refinement of the predictions with minimum data input on place-level or site characteristics for what we refer to as “hyper-localization”.

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Keywords DSSAT · GSSAT · Spatial simulation · Fertilizer recommendation · Hyper-localization

Introduction

The global population is projected to reach approximately 10.1 billion by 2050. Sub-Saharan Africa (SSA) is expected to experience the most significant population growth (United Nations 2019), reaching 2 billion by 2050 (International Monetary Fund 2024), up from the current 1.2 billion (World Population Review 2024). This will exert immense pressure on farming systems to increase agricultural productivity and production to meet the rising food demand from the growing population. Evidence (Lipper et al. 2014; Canning et al. 2015; van Ittersum et al. 2016) shows that cereal production in SSA will need to increase by 60–80% to meet the projected threefold rise in cereal demand. In the absence of supportive interventions, achieving this increase in production may not be possible and could be further hampered by other challenges, such as climate change, rising costs of inputs, including fertilizer and seed, and soil fertility degradation.

Thus, farming systems in SSA, most of which are maize (*Zea mays* L.) based, will need sustainable transformation to ensure productivity and production improvement in the face of current and future challenges. In Zambia, maize is the main staple food, covering approximately 1.4 million hectares (Food and Agriculture Organization of the United Nations [FAO] 2020), and it is crucial for food security and livelihoods. However, smallholder farmers in Zambia face significant challenges in achieving optimal maize yields, primarily due to inefficient nutrient management and climate variability, owing to the heavy reliance on rainfed production. Despite the extensive maize growing area, actual yields in Zambia fall short of attainable yields due to inefficient fertilizer use, nutrient deficiencies, and adverse weather conditions (Chikowo 2016).

Climate variability and extremes further intensify these challenges, with rising temperatures and unpredictable rainfall patterns impacting maize production (Zhao et al. 2017), resulting in increases in global hunger (FAO 2018). These extreme weather events are common in Zambia. For example, the 2015–2016 El Niño event caused severe drought conditions in

Zambia, drastically reducing maize yields and highlighting the vulnerability of smallholder systems to climatic extremes (Beyene et al. 2016). In the 2023–2024 production season, Zambia faced another El Niño event that resulted in severe drought, affecting nearly 1 million hectares of the 2.2 million hectares of planted maize (Lusaka Times 2024). The 2023–2024 El Niño-induced droughts resulted in a yield reduction of over 60%, from an average of 1.72 metric tons per hectare (mt ha⁻¹) in the 2022–2023 season to 0.68 mt ha⁻¹ (Ministry of Agriculture 2024).

Unlike most areas of the world, crop yields have not increased substantially in SSA, and therefore, a large share of the population in this region faces hunger (Arouna et al. 2021; FAO et al. 2022). The Green Revolution successfully boosted crop productivity with high-yielding varieties, increased fertilizer application, and better water and pest management in Asia, Europe, and North America, contrasting greatly with the current situation in Zambia (Gu and Yang 2022). Agricultural advisory services in SSA have been promoting blanket fertilizer recommendations, in which a single fertilizer type and rate is used for large but agroecologically heterogeneous areas. Fertilizer nutrient application rates in SSA average only about 16 kg ha⁻¹ per year according to IFASTAT (2019), compared with over 100 kg ha⁻¹ in Europe and North America and over 150 kg ha⁻¹ in China. Although the average fertilizer use in Zambia has surpassed the 50 kg ha⁻¹ required by Abuja declaration (Vanlauwe and Dobermann 2020), reaching 63.9 kg ha⁻¹ in 2021 according to World Bank data, fertilizer rates of 200 kg ha⁻¹ of N-P₂O₅-K₂O-S 10–20–10–6, also referred to as D Compound, as basal application and 200 kg ha⁻¹ of urea as topdressing have been recommended for maize production across agroecologies. Blanket fertilizer recommendations often fail to address spatial variability in soil fertility and crop nutrient needs, leading to inefficient fertilizer application, limited productivity, and low profitability, thus resulting in a decreased demand for fertilizer (Chapoto et al. 2016). For instance, uniform fertilizer recommendations have led to underperformance in regions with diverse soil types and nutrient profiles (Hagglblade et al. 2019; Rurinda et al. 2020). This underscores the need for a more tailored approach to nutrient management.

One promising alternative is the site-specific nutrient management approach, a strategy for feeding crops with nutrients as and when needed. Site-specific nutrient management optimizes fertilizer application by considering location-specific soil, climate, and socio-economic conditions (Chivenge et al. 2022; Dobermann et al. 2002, 2004; Buresh et al. 2019). Several studies have shown that application of site-specific nutrient management principles improved nutrient use efficiency (NUE), crop yield, and farmers' profits (Shankar et al. 2014; Dutta et al. 2020; Timsina et al. 2021; Chivenge et al. 2022).

Crop modeling has been widely used for addressing these challenges, providing tailored nutrient management recommendations. Crop models embedded within shells of the Decision Support System for Agrotechnology Transfer (DSSAT; Jones et al. 2003), the Agricultural Production Systems Simulator (APSIM; Keating et al. 2003), and the Quantitative Evaluation of the Fertility of Tropical Soils (QUEFTS; Janssen et al. 1990) simulate crop growth under varying conditions, offering insights into optimal fertilizer strategies. However, unlike these models that provide point-based simulations, the Geographic Support System for Agrotechnology Transfer (GSSAT2) effectively handles spatial simulations. Leveraging web and cloud technologies, GSSAT2 is designed to enable large-scale simulations and provide spatially informed recommendations for optimizing crop management (Pavan et al. 2023; Bombana 2023).

This study is the first evaluation of the performance of GSSAT2 in assessing crop yields, specifically for maize in Zambia. By assessing yield potential, yield gaps, and spatio-temporal variability of nutrient requirements using GSSAT2, this study aims to improve crop growth simulation to inform fertilizer application strategies for smallholder farmers and maximize yields and farm profits. This will help enhance maize productivity and production in Zambia and set a precedent for applying similar approaches to other crops and regions.

Materials and methods

Overview of DSSAT: CERES-maize

Crop simulations in GSSAT2 are handled through DSSAT Cropping System Models. In this study, we used the Crop Environment Resource Synthesis (CERES)-Maize model embedded within DSSAT (v4.8.2). CERES-Maize is a deterministic model designed to simulate the effects of rainfed and irrigated conditions, planting dates, and various management practices and environmental variables for maize growth and yield (Hoogenboom et al. 2019). This model accounts for crop development in relation to these conditions, including leaf area growth, daily plant growth rate, biomass accumulation, and the partitioning of biomass to different plant organs. The CERES-Maize model captures major physiological processes, such as photosynthesis, respiration, biomass accumulation, and carbon partitioning. Photosynthesis is simulated using the radiation use efficiency method, while evapotranspiration is modeled using the Priestley–Taylor equation (Jones et al. 2003; Hoogenboom et al. 2019). The maize cultivar file contains six user-adjustable parameters related to growth (Table 1). Phenological development is driven by growing degree days (GDD), or thermal time, and photoperiod. GDD are calculated based on daily maximum and minimum temperatures, and in some growth stages, day length is also considered (Jones and Kiniry 1986; Jones et al. 2003). The potential growth of maize in the CERES-Maize model depends on photosynthetically active radiation and its interception.

Overview of GSSAT2

GSSAT2, the latest iteration of the legacy GSSAT, leverages cutting-edge web and cloud technologies to enhance spatial crop modeling and decision support capabilities (Pavan et al. 2023; Bombana 2023). This advanced version allows users to perform large-scale simulations by defining specific areas of interest or by uploading geographic information system (GIS)-compatible input files that delineate the study region. It also includes a feature to select a single location using latitude and longitude information and define site or field characteristics, including previous crop grown, crop residue management, and initial soil properties.

Table 1 Genetic coefficients in the CERES-Maize model

Genetic coefficient	Definition	Value*
P1	Thermal time from seedling emergence to the end of the juvenile phase (expressed in degree days above a base temperature of 8 °C), during which the plant is not responsive to changes in the photoperiod	240
P2	Extent to which development (expressed as days) is delayed for each hour increase in the photoperiod above the longest photoperiod at which development proceeds at a maximum rate (which is considered to be 12.5 h)	0
P5	Extent to which development (expressed as days) is delayed for each hour increase in photoperiod above the longest photoperiod at which development proceeds at a maximum rate (which is considered to be 12.5 h)	770
G2	Maximum possible number of kernels per plant	550
G3	Kernel filling rate during the linear grain filling stage under optimum conditions (mg per day)	9
PHINT	Phyllochron interval: the interval in thermal time (degree days) between successive leaf tip appearances	55

*Source: Corbeels et al. (2016)

GSSAT2 excels in assessing crop yield gaps in targeted areas, with consideration of the spatial variability of soil, weather, and management practices. This geospatial decision support tool is particularly valuable for precision agriculture, where tailored recommendations are essential for optimizing inputs and improving crop yields.

One of the key features of GSSAT2 is its ability to efficiently analyze large datasets through parallel processing, which facilitates completion of comprehensive assessments in a time-efficient manner. The system integrates diverse input data, including weather, soil, and crop management information, into a unified grid format, thereby providing a detailed and flexible platform for crop and nutrient decision-making (Pavan et al. 2023). More recently, sensitivity analysis and economic analysis modules have been added to assess the optimum planting date and the optimal fertilizer use by farmers on a spatial scale.

Another key feature of GSSAT2 is its integration of the DSSAT-Pythia tool, which simplifies the usage of the DSSAT Cropping System Model in geospatial contexts by facilitating the mass execution and the linking of GIS soil and weather data. In addition, it provides perks like parallelization support and basic data analytics (Bombana 2023; Hoogenboom et al. 2019; Joshi et al. 2021).

Country focus

This study utilized maize data from Zambia. The geographical position of Zambia, mostly lying

between latitudes 8° and 18° S and longitudes 22° and 34° E, at the crossroad of Central, Southern, and Eastern Africa, confirms its representativeness of a wide range of agroecological zones for maize production within the southern African region. The climate across Zambia is tropical, mostly humid subtropical or tropical wet and dry (Beck et al. 2018). Precipitation amounts range from 500 to 1400 mm per year during the rainy season, spanning from November to April, followed by a dry season between May and October. Average temperature is around 30 °C during the summer period but can drop to 5 °C during the cold period between May and August.

Model parameterization, calibration, and testing

The commercial hybrid maize variety SC513, an early maturing variety, was used to calibrate the maize model, with parameters obtained from Corbeels et al. (2016) (Table 1). Data used for the model calibration were obtained from field experiments conducted by the International Maize and Wheat Improvement Center (CIMMYT) in Monze (latitude 16° 14' 24" S, longitude 27° 26' 24" E, elevation 1103 m.a.s.l.) from 2005 to 2011. Model testing using these parameters showed root mean square error (RMSE) values of 4%, 3.4%, 10%, and 10% for simulating silking date, maturity date, grain yield, and vegetative aboveground biomass yield, respectively (Corbeels et al. 2016).

Testing GSSAT using maize trials in Monze, Zambia

Input data for the simulations included the calibrated crop parameters as well as weather, soil, and crop management data. Weather data comprised measured daily satellite data on precipitation, solar radiation, and minimum and maximum temperatures, downloaded from the National Aeronautics and Space Administration (NASA) Langley Research Center (LaRC) Prediction of Worldwide Energy Resources (POWER) project, funded through the NASA Earth Science/Applied Science Program. Soil data were downloaded from the SoilGrids database (Poggio et al. 2021) using the longitudinal and latitudinal values of Monze (Table 2). Then, the hyper-localization features of GSSAT2 were used to overwrite the SoilGrids data with soil properties measured at the site in Monze (Table 3). The simulations were run first to test the simulation outputs for Monze covering 38 years (1985–2016), including the years the field experiments were conducted.

Crop management information consisted of November 15 sowing of maize seeds at a plant population of 44,444 ha⁻¹, with basal application of 165 kg ha⁻¹ of N-P₂O₅-K₂O-S 10–20–10–6 during planting and topdressing with split application of 200 kg ha⁻¹ of urea at Week 5 (one-half) and Week 7 (one-half) after sowing, as per Corbeels et al. (2016). Land preparation and crop management were optimal, with weed control by a pre-emergence application of glyphosate (*N*-(phosphonomethyl) glycine, 41% active ingredient) at a rate of 3 L ha⁻¹ followed by regular hand-weeding as necessary to keep the plots free of weeds.

Simulated yields were compared with measured yields obtained in the field experiment. Yields simulated with and without enabling the soil analysis feature were also compared to assess the hyper-localization or customization effect of the impact.

Assessing crop growth and optimizing nutrient management using GSSAT2

After testing GSSAT2 in Monze, we proceeded with running GSSAT2 spatially across Zambia over 38 years (1985–2023) of daily weather data. The same input data for Zambia, including automatically downloaded satellite weather from NASA

POWER and SoilGrids and user-defined crop management data, were used. Crop management consisted of planting two maize seeds per hole at a spacing of 0.75 m × 0.25 m, thinning to one plant per hole after emergence, and applying 165 kg ha⁻¹ of D Compound (N-P₂O₅-K₂O-S 10–20–10–6) during sowing (basal application) and topdressing with urea at (1) 100 kg ha⁻¹ (total N fertilizer application of 63 kg N ha⁻¹ [treatment N63]), (2) 200 kg ha⁻¹ (total N fertilizer application of 109 kg N ha⁻¹ [treatment N109]), and (3) 300 kg ha⁻¹ (total N fertilizer application of 155 kg N ha⁻¹ [treatment N155]). To optimize the nutrient supply to the plant and minimize nutrient losses, the topdressing was applied at Week 5 (100%) for N63; at Week 5 (one-half) and Week 7 (one-half) for N109; and at Week 5 (one-third), Week 6 (one-third), and Week 7 (one-third) for the N155 treatment. Land preparation and crop management were assumed optimal with recommended weed and disease management.

Five different planting dates were evaluated for 38 years across Zambia: October 15, November 15, December 15, January 15, and February 15. Maize growth and yield were simulated for different fertilizer rates—no fertilizer, N63, N109, and N155 – using these dates to determine the best planting date. Yield gap was also estimated as the difference between the rainfed potential yield (water-limited yield) and the water- and N-limited yield (without N fertilizer application). The rainfed potential yield assumes nutrients (including phosphorus [P] and potassium [K]) are fully supplied to the plant when needed, and thus, no nutrient stress occurs.

A simple partial budget economic analysis considering input costs (N) and output price (maize grain), as shown in Table 4, was used to determine the most efficient fertilizer rate, the rate with the highest marginal return on investment. No other costs were considered. The return on investment was obtained as the difference between income (grain yield × unit price of grain) and the total cost of fertilizers (quantity × unit cost) and seeds (plant population × unit seed weight × unit cost). For the maize grain price and the fertilizer cost combinations, four scenarios were considered: high maize price at low and high fertilizer cost and low maize price at low and high fertilizer cost.

Table 2 Snap of the SoilGrids profile* nearest to Monze

* ZM05506170										Soilgrids + HC27									
ZMB		SandClayL		200	ISRIC														
@SITE	COUNTRY	LAT	LONG	SCSFamily															
–99	ZM	–16.208	27.542	HC_GEN0011															
@	SCOM	SALB	SLU1	SLDR	SLRO	SLNF	SLPF	SMHB	SMPX	SMKE									
	BK	0.1	6	0.5	75	1	1	SA001	SA001	SA001									
@	SLB	SLMH	SLLL	SDUL	SSAT	SRGF	SSKS	SBDM	SLOC	SLCL	SLSI	SLCF	SLNI	SLHW	SLHB	SCEC			
	5	A	0.162	0.27	0.399	1	0.56	1.45	1.58	27.03	22.43	–99	0.12	5.98	–99	15.4			
	15	A	0.173	0.283	0.403	0.85	0.45	1.47	1.34	28.9	21.66	–99	0.09	6.06	–99	13.4			
	30	AB	0.189	0.301	0.409	0.7	0.33	1.5	1.02	31.58	20.35	–99	0.07	6.16	–99	13.1			
	60	BA	0.204	0.317	0.415	0.5	0.24	1.55	0.65	34.06	19.14	–99	0.06	6.27	–99	13.6			
	100	B	0.203	0.315	0.414	0.38	0.25	1.61	0.38	33.93	18.65	–99	0.05	6.41	–99	13.8			
	200	BC	0.194	0.303	0.409	0.05	0.3	1.66	0.21	32.37	18.53	–99	0.05	6.58	–99	13.7			

*Here is the description of the codes used in the Table 2: ZM05506170 for the name of the profile in SoilGrid database; ZMB for Zambia, SandClayL for sandy clay loam (texture); ISRIC for World Soil Information Centre; SCOM for color (moist, Munsell hue); SALB for Albedo (fraction); SLU1 for Stage 1 evaporation limit (mm); SLDR for Drainage rate (fraction day^{–1}); SLRO for runoff curve number; SLNF for mineralization factor (0 to 1 scale); SLPF for soil fertility factor (0–1 scale); SMHB for pH in buffer determination method (code); SMPX for phosphorus determination (code); SMKE for potassium determination (code); SLB for depth (base of layer, cm); SLMH for master horizon; SLLL for lower limit of plant extractable soil water (cm³ cm^{–3}); SDUL for drained upper limit of plant extractable soil water (cm³ cm^{–3}); SSAT for upper limit (saturated, cm³ cm^{–3}); SRGF for root growth factor (0–1); SSKS for saturated hydraulic conductivity (macropore, cm h^{–1}); SBDM for bulk density (moist, g cm^{–3}); SLOC for organic carbon (%); SLCL for clay (<0.002 mm, %); SLSI for silt (0.05–0.002 mm, %); SLCF for coarse fraction (> 2 mm, %); SLNI for total nitrogen (%); SLHW for pH in water; SLHB for pH in buffer; SCEC for cation exchangeable capacity (cmol kg^{–1})

Table 3 Soil data used for GSSAT2 simulation when the soil analysis feature was enabled for some key site characteristics (place-level inputs or hyper-localization data)

Soil layer (cm)	Organic C (%)	pH in water	Bulk density (cm ³ cm ⁻³)
0–22	0.6	4.4	1.67
23–56	0.3	5.1	1.48
57–80	0.05	5.3	1.48
81–107	0.05	5.6	1.33
> – 107	0.05	5.7	1.48

Organic C, pH in water and bulk density correspond to SLOC, SLHW and SBDM, respectively in Table 2

Source: Corbeels et al. (2016)

Results

Calibrated maize yields in Monze, Zambia

The maximum yield used for running the Generalized Likelihood Estimator (GLUE) for calibration of the maize cultivar in Monze was 6.38 Mg ha⁻¹ (Corbeels et al. 2016). When the place-level input was enabled

to consider the same soil information, annual maize yield in Monze ranged from 1.2 Mg ha⁻¹ for no fertilizer to 8.2 Mg ha⁻¹ for N109, with mean values of 2.4 Mg ha⁻¹ and 6.7 Mg ha⁻¹, respectively (Table 5). When the place-level input was not enabled, higher yields were achieved, ranging from 1.9 Mg ha⁻¹ for no fertilizer to 8.3 Mg ha⁻¹ for N109, with mean values of 4.2 Mg ha⁻¹ and 6.7 Mg ha⁻¹, respectively. Yield achieved with N109 was close to the rainfed potential yield.

Maize yield potential and best planting period

Maize yield varied spatially from 0 kg ha⁻¹ to 13.2 Mg ha⁻¹ (Table 6) across Zambia depending on the planting date. The highest yield was achieved with February planting and the lowest with October planting. Planting in October and February would be, however, riskier, respectively in the eastern and the western part of the country, with a wider range of lower yields compared to planting in November and December (Fig. 1A). The cumulative probability distribution of grain yield across years also shows that in

Table 4 Average market price of inputs and farm-gate price of maize in 2024

Item	Zambian kwacha (ZMW)	U.S. dollar (U.S. \$)
Urea price (50 kg)	900.00	34.10
NPK-S (D compound) price (50 kg)	750.00	28.42
N cost—high (based on urea price)	39.10	1.48
N cost—low (subsidized to 250 ZMW 50 kg ⁻¹)	10.90	0.41
Maize grain price (1 mt)—high	9000.00	341.04
Maize grain price (1 mt)—Table 5 Maize yield simulated for Monze with and without place-level input enabled	6600.00	250.09
low		
Maize seed price (1 kg)	70.00	2.65

Exchange rate: 1.00 U.S. dollar = 26.39 Zambian kwacha

Table 5 Maize yield simulated for Monze with and without place-level input enabled

Soil analysis enabled	Treatments	Simulated yields (kg ha ⁻¹)					
		Min	Median	Mean	Max	2.5th Percentile	97.5th Percentile
Yes	No fertilizer	1170	2511	2397	3466	1179	3193
	N109	5473	6786	6743	8176	5816	8151
No	No fertilizer	1924	4335	4200	5624	1950	5569
	N109	5473	6786	6749	8276	5816	8270
Difference	No fertilizer	– 754	– 1824	– 1803	– 2158	– 771	– 2376
	N109	0	0	– 6	– 100	0	– 119
	Rainfed potential	5473	6786	6752	8276	5816	8271

Table 6 Summary descriptive statistics of simulated maize yields for different planting dates

Planting dates	Simulated yields (kg ha ⁻¹)					
	Min	Median	Mean	Max	2.5th percentile	97.5th percentile
15-Oct	0	6429	4641	10,829	0	8885
15-Nov	0	7172	7039	11,534	4344	8845
15-Dec	0	7310	7186	12,636	3048	9696
15-Jan	0	7235	6568	12,958	0	10,861
15-Feb	0	5441	4437	13,160	0	9285

more than 25% of the cases, planting in October and February would lead to crop failure (no grain yield, Fig. 1B), probably occurring in the eastern and western zones, respectively. Total season rainfall is also lower for October followed by February (Fig. 1C). In the other part of the country, planting in these months can result in reasonable yields as depicted in Fig. 1B in more than 50% of the cases.

A minimum yield of 0 kg ha⁻¹, associated with crop failure in dry years (prolonged droughts), was evident across the planting dates, but higher proportion for October, February and January compared to November and December. Applying 2.5th and 97.5th percentiles on the simulated yields showed that the November and December planting dates were less risky for maize production, as crop failure occurrence during this time was minimal at less than 2.5%.

Maize yield gap and response to fertilizer application

Gaps between the rainfed potential yields and the yields achieved without N fertilizer application could exceed 4 Mg ha⁻¹ (Table 7, Figs. 2A and Fig. 2B). Application of N fertilizer (63 kg ha⁻¹, 109 kg ha⁻¹, and 155 kg ha⁻¹) increased yields by 134%, 156%, and 156%, respectively, thereby narrowing the yield gaps (Table 7, Fig. 2B).

Fertilizer requirements for maize production

Fertilizer requirements for profit optimization varied spatially depending on the planting date and the variations in maize grain price and fertilizer cost (Figs. 3 and 4). For early season planting (November 15), at both low and high maize prices, N109 and N63 were deemed the most appropriate treatments to be recommended at a high fertilizer cost (Fig. 3A, B), while N109, N155, and N63 were the most appropriate treatments at a low fertilizer cost, depending on the

location (Fig. 3C, D). Low fertilizer costs allow the recommendation of higher profitable fertilizer rates. The same trend was observed for late planting (February 15), where the no-fertilizer treatment was the most efficient recommendation in the western part of the country (as planting in February was not advisable in that region as per Fig. 1), versus N63 and N109 at a high fertilizer cost in the remaining part of the country (Fig. 4A, B); high fertilizer rates (N155, followed by N109) were the most efficient treatments in many areas at low fertilizer costs (Fig. 4C, D).

Discussion

This study showed that GSSAT2 can operate at different scales, both locally (field level), using single geographic coordinates, and regionally (village, district, region, and country levels), with spatial simulations. Simulated yields for Monze fell within the range of measured yields that were used for the calibration (Corbeels et al. 2016). The reduction in yield obtained by switching on the soil analysis feature in GSSAT2 (utilizing on-site soil analysis data) was due to the lower carbon content of the local soil data of Monze (0.6% in the topsoil and 0.05% in the subsoil) compared to the carbon content of the soil from Soil-Grids data (1.58% in the topsoil and 0.21–0.65% in the subsoil) used as the default values in GSSAT2. This highlights the importance of using locally available soil management information to customize fertilizer recommendations.

This study also confirmed that maize yields vary in time and space in Zambia. The rainfed potential yield with no nutrient limitation ranged from 0 to 13 Mg ha⁻¹, depending on time and space (Fig. 1). During the years of droughts, complete crop failures occurred, with zero grain yields. Crop failures were more pronounced when planting too early (October,

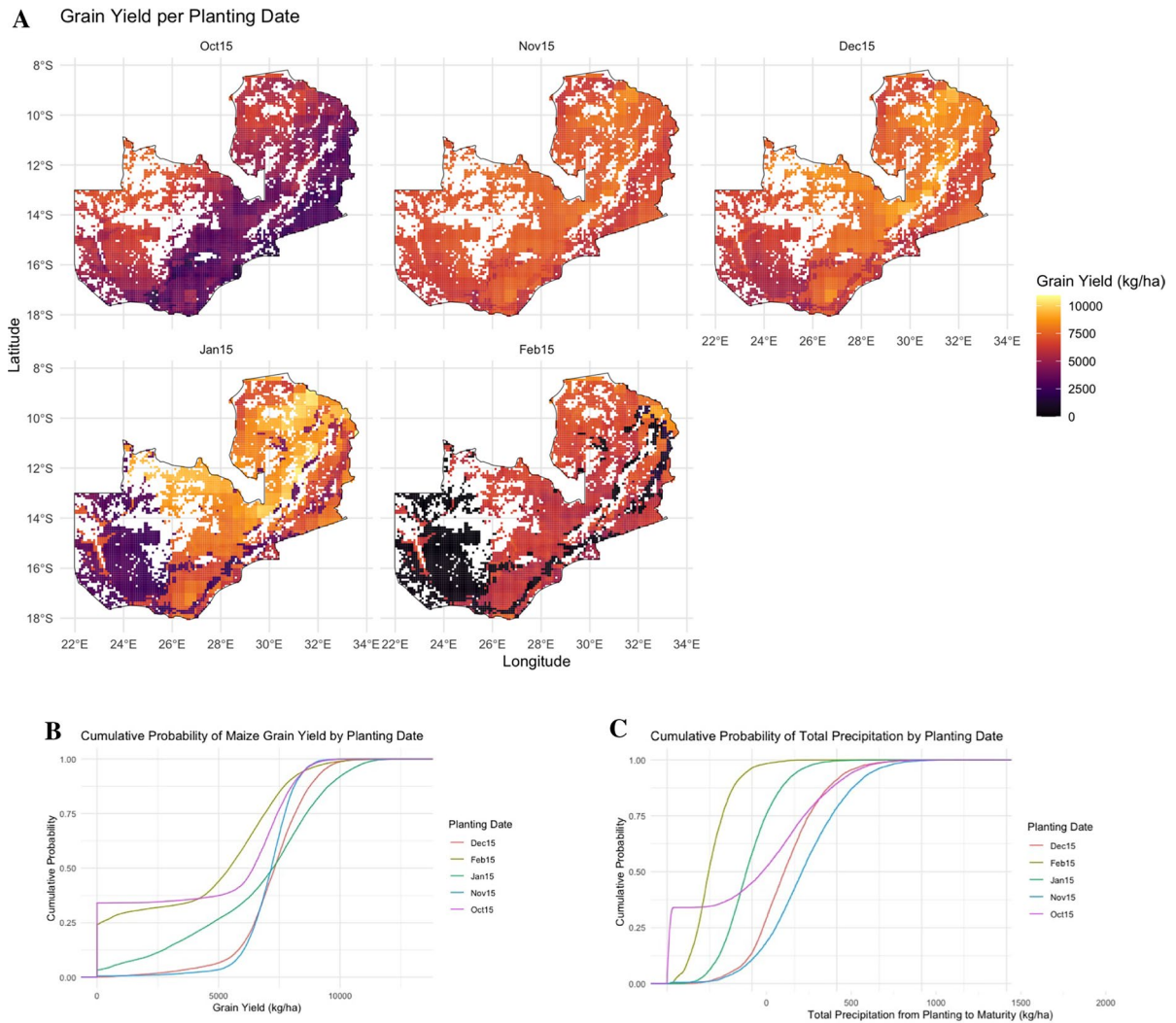


Fig. 1 Spatial distribution of maize average rainfed potential yield (kg ha^{-1}) and total precipitation (mm) for different planting dates from mid-October through mid-February over 38 years (1985–2023) in Zambia. **A** compared grain yield spatial distribution for the 5 planting dates: Yields varied spatially

and ranged from low (red) to high (yellow). **B** shows the cumulative probability distribution of grain yield for different planting dates. **C** shows the cumulative probability distribution of total precipitation accumulated from planting to maturity for different planting dates

Table 7 Summary descriptive statistics of yield response to nitrogen fertilizer application

Treatments	Simulated yields (kg ha^{-1})					
	Min	Median	Mean	Max	2.5th percentile	97.5th percentile
No fertilizer	0	2796	3022	9840	893	6758
N63	0	6544	6374	9840	3599	8169
N109	0	7157	6999	9921	4297	8637
N155	0	7169	7031	11,259	4297	8805
Rainfed potential	0	7172	7039	11,534	4344	8845

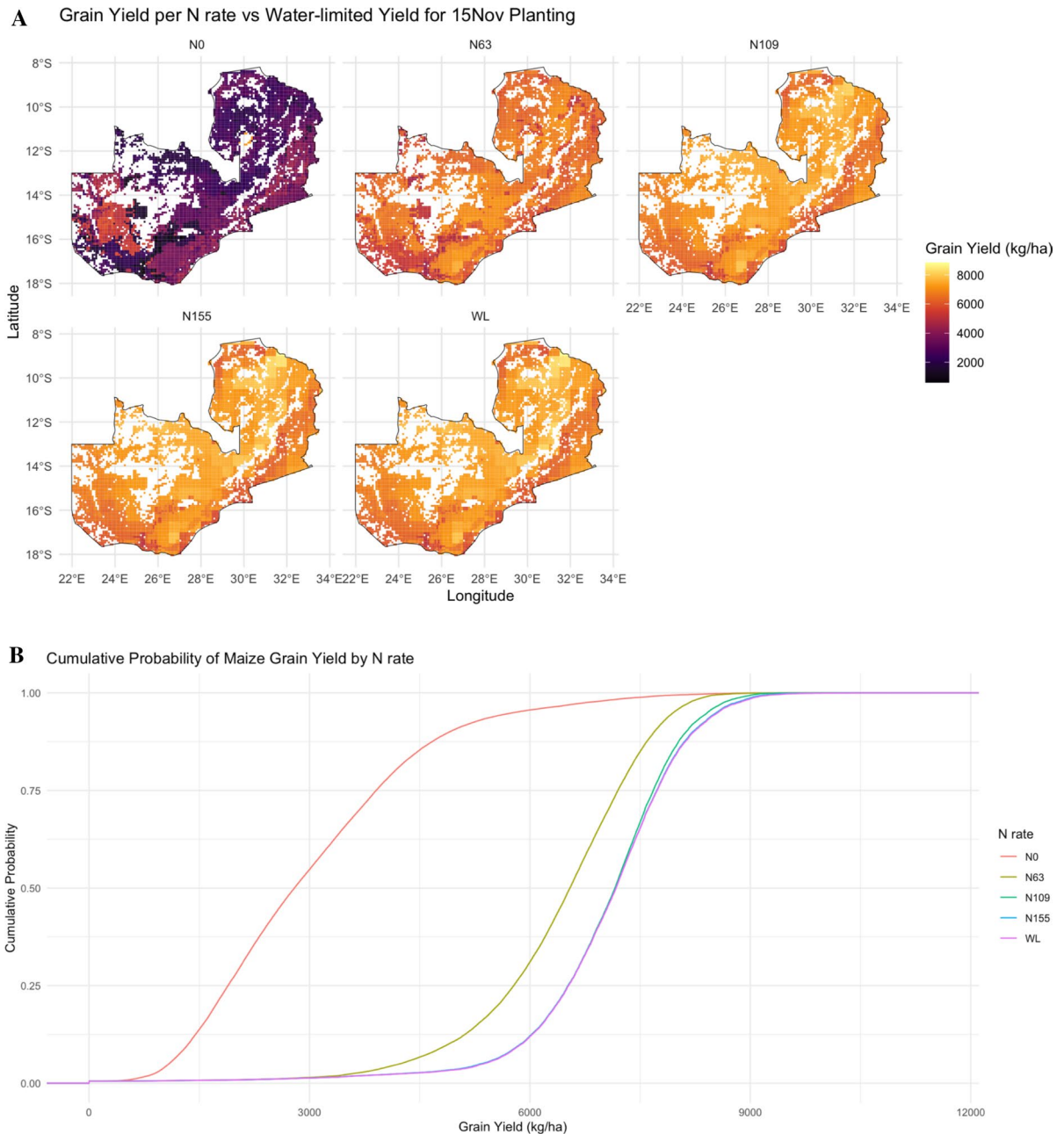


Fig. 2 Spatial distribution (A) and cumulative probability of average maize yields as affected by nitrogen fertilizer application compared to rainfed potential yield for November planting across 38 years. In A, Yields varied spatially and ranged from

low (dark pink) to high (yellow). In B, WL stands for water-limited yield or rainfed potential yield; N0, N63, N109 and N155 stand for nitrogen applications are the rates of 0, 63, 109 and 155 kg ha⁻¹

especially in eastern Zambia) or too late (January onward for the western Zambia) (Fig. 1, Table 6), emphasizing the critical importance of ensuring the planting date is within the November to December

window to secure reasonable yields. However, early planting in October can occur in western Zambia, and late planting in January–February in eastern Zambia. These results nuance slightly the previously suggested

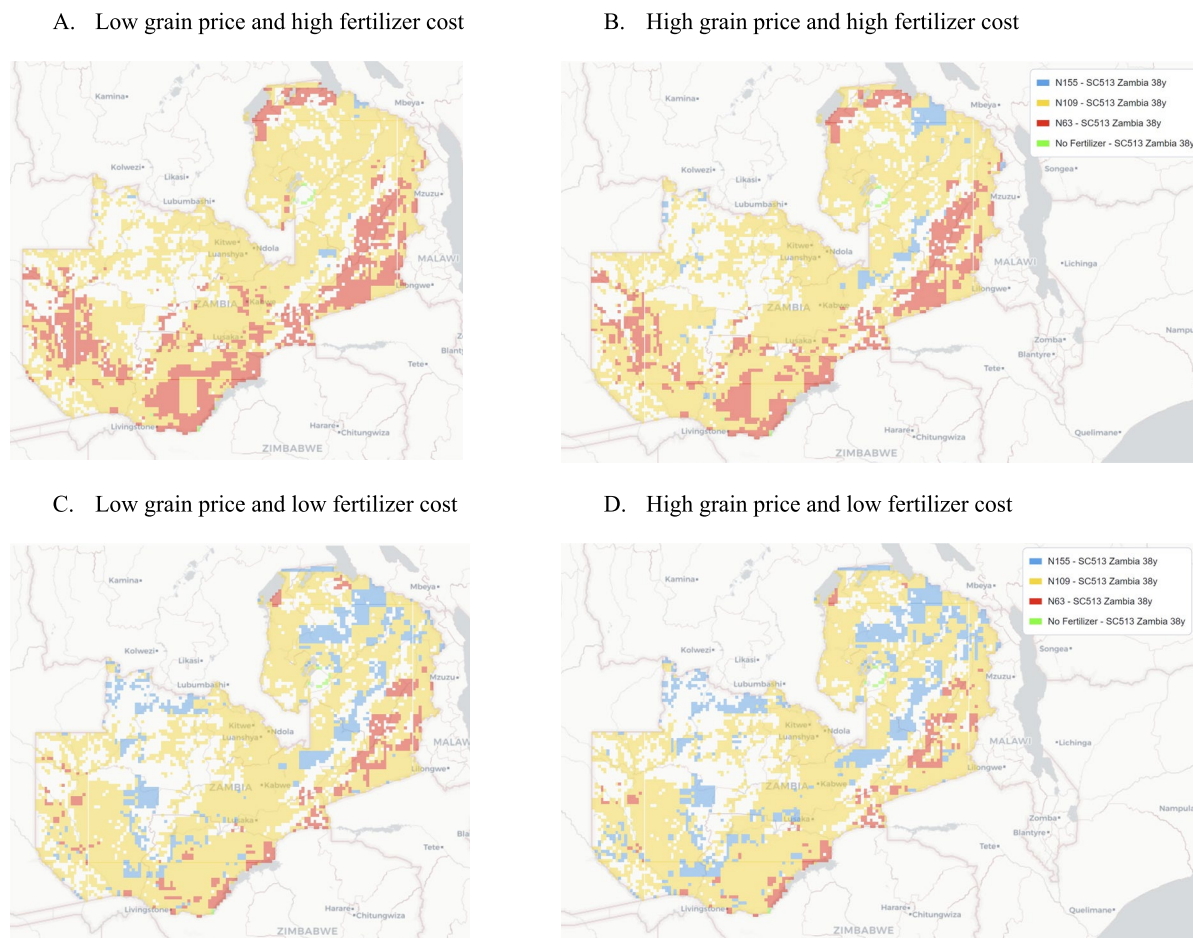


Fig. 3 Spatial variability of fertilizer requirements as affected by maize grain price and fertilizer cost for early season planting (November 15)

range of planting, i.e., beginning of October to end of December (Waldman et al. 2019), which could be attributed to localized climatic and soil conditions in the study area. The findings highlight the practical need for farmers to adhere to the optimal planting window and suggest that agricultural policies should support this through improved forecasting and extension services.

Huge gaps in maize yields were estimated for Zambia, reflecting spatial variations. With a median rainfed potential yield of 7.2 Mg ha^{-1} for November planting versus 2.8 Mg ha^{-1} with no fertilizer application, the yield gap due to nutrient supply deficit could amount to 4.4 Mg ha^{-1} (Table 7). This gap was reduced with the application of 63 kg, 109 kg, and 155 kg N ha^{-1} , thereby increasing yields by

134%, 156%, and 156%, respectively. The more than 100% yield response to N supply implies that external N application is necessary to reduce the yield gaps in Zambia. These findings align with other studies in SSA, such as Folberth et al. (2013), which demonstrated that increased nutrient supply and improved cultivars could raise yields to 7 Mg ha^{-1} . This suggests that current farming practices have yet to fully harness Zambia's yield potential, underscoring the need for enhanced nutrient management to optimize maize productivity. External nutrient application can be achieved by applying high amounts of organic resources with a lower N content or a combination of organic resources and mineral fertilizers with a higher N content in lower amounts to ensure greater efficiency of the mineral

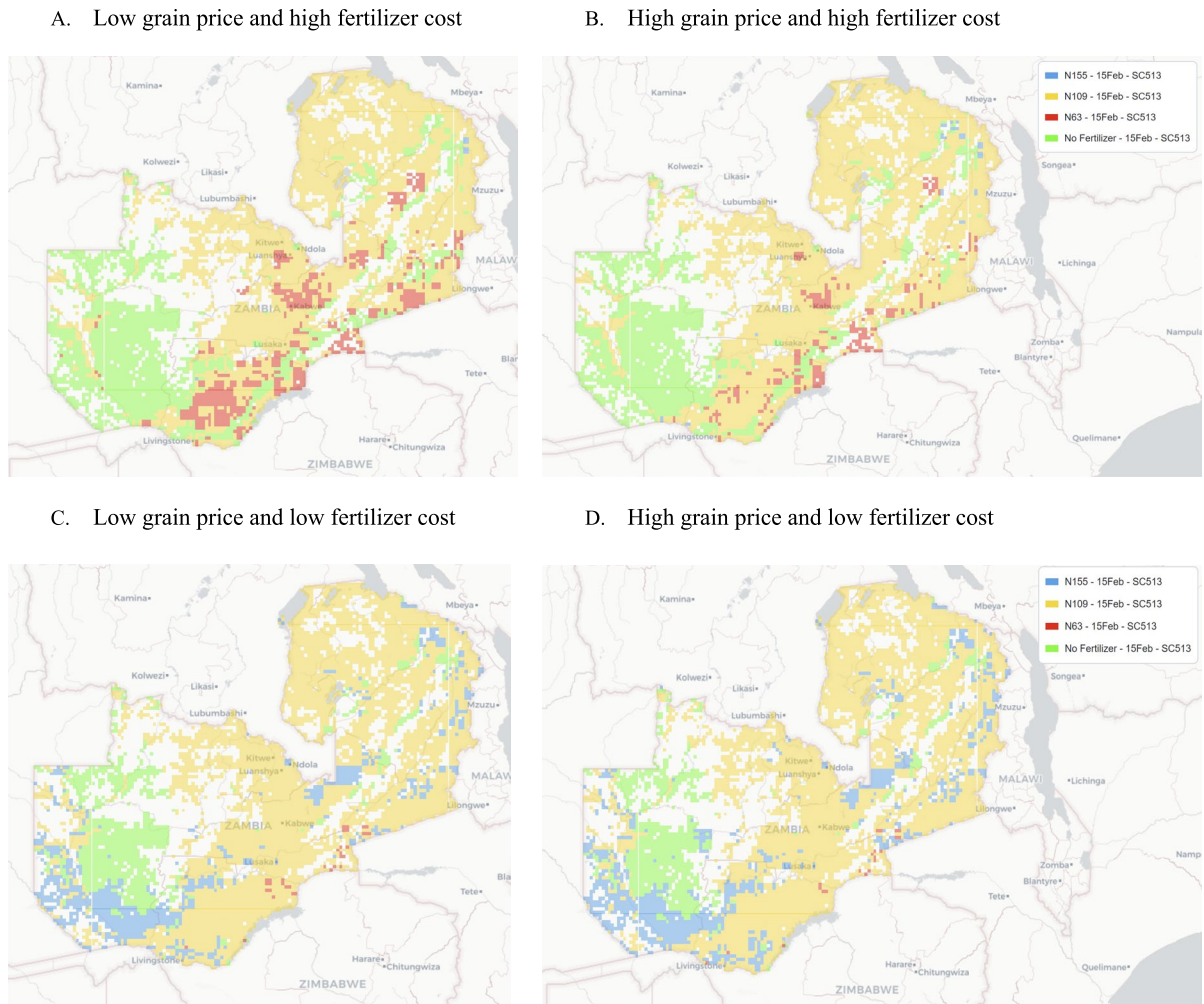


Fig. 4 Spatial variability of fertilizer requirements as affected by maize grain price and fertilizer cost for late season planting (February 15)

fertilizers (Timsina 2018). This can be done following good agronomic practices and the 4R Principles of Nutrient Stewardship, involving the application of the right source of nutrient, at the right place, at the right time, and at the right rate for enhanced efficiency (Johnston and Bruulsema 2014), which is in line with GSSAT2 nutrient recommendations. Splitting high N rates into two to three applications is also advisable (Ma et al. 2021; Hu et al. 2021; Timsina et al. 2021; Varshini and Babu 2021). Closing the yield gap will also require robust extension and advisory systems that ensure the efficient flow of information from research to the farmer and vice versa.

A simple economic analysis considering different scenarios of maize grain price and fertilizer cost in the selection of the most efficient fertilizer rates among the 0, 63 kg, 109 kg, and 155 kg N ha⁻¹ treatments showed that blanket rates are not profitable in all locations (Figs. 3 and 4). The most efficient fertilizer rate varied spatially, confirming the importance of tailored fertilizer recommendations compared to blanket recommendations (Dutta et al. 2020; Timsina et al. 2013). Although the N109 treatment (165 kg of D Compound, followed by 200 kg ha⁻¹ of urea) seemed efficient in most cases, lower fertilizer rates (N0 and N63) were more efficient in some places when fertilizer costs were high and higher fertilizer

rates (N155) were recommended when fertilizer costs were lower. These results suggest that fertilizer cost is a major driver of fertilizer recommendations, as it affects the profit margin.¹ Hence, maintaining fertilizer price at a reasonable level is advisable to encourage farmers to use enough nutrients to avoid depleting the soil while preventing an oversupply of nutrients. The recommendation for high N application also matched areas with high yield potential (Figs. 2 and 3), unlike low production areas, highlighting the importance of adjusting fertilizer recommendations to the responsiveness of the location to fertilizer application. It is noteworthy that the highest N fertilizer rate (N155) was applied in three splits. This practice is advised for areas where high rates are recommended to match N supplies with plant nutrient demands and reduce N losses into the environment. However, the profitability of an extra application cost should also be evaluated.

The GSSAT2 simulations also demonstrated the importance of adjusting fertilizer recommendations based on planting time. Late sowing came with a higher risk of crop failure and lower yield due to drought or insufficient soil moisture. Hence, lower fertilizer rates were deemed more practical and useful than higher rates in some locations.

Overall, GSSAT2 appears to be a solid tool for estimating maize yield potential and yield gaps and for optimizing nutrient management in space and time. However, validation of the results of this study is recommended with wider testing across different agro-ecologies and socioeconomic settings. In addition, the settings apply to the whole geographic area selected for the simulation with NASA POWER-based weather spatial resolution of 0.5° (about 55.5 km at the equator) $\times$ 0.625° for most parameters, except for solar parameters ($1^\circ \times 1^\circ$), and soil grid resolution of 5 arc-minute grid (about 9.3 km at the equator), place-level inputs can be used to further adjust, or hyper-localize, the recommendations to farmers' specific conditions (Table 5). Besides

locally available soil analysis results, place component inputs can include information on the previous crop, crop residue management, and initial soil moisture. These variables were not included in the current analysis with the countrywide simulations. Yet, such useful features of GSSAT2 for hyper-localizing fertilizer recommendations are a work in progress and will be used in our upcoming research with updated data and information. Moreover, the current version of GSSAT2 does not cover costs associated with transport and labor besides fertilizer application, which limited the economic analysis. Considering these extra economic variables could have increased the production cost, reduced the profit margin and led to recommending lower rates, especially in low production areas. Further improvements in GSSAT2 will include more economic variables for more realistic profit analysis and integrate higher resolution weather data, especially for rainfall, which is important in the agricultural context of SSA.

Although the results of this analysis are highly encouraging, especially regarding generating optimal location-specific nutrient management practices, we suggest that GSSAT2 recommendations be used to guide validation trials in the field to further improve the model engine and generate more robust results for the delivery of reliable soil fertility management advisories to farmers.

Conclusion

This study evaluates GSSAT2, a spatial simulation tool based on DSSAT crop modeling, for optimizing maize nutrient management in Zambia. GSSAT2 as a decision support tool offers flexibility by operating at different scales from field level (individual farmer) to regional/national scale, enabling both localized recommendations and broader planning across different levels. GSSAT2 modeling results also showed that substantial yield gaps exist in Zambia maize production, with median rainfed potential yields of 7.2 Mg ha^{-1} achievable through proper nutrient management compared to 2.8 Mg ha^{-1} without fertilizer, representing a yield gap of 4.4 Mg ha^{-1} . Also, our simulations show the nitrogen application rates of 63–155 kg N ha^{-1} increased yields significantly (> than 100%) demonstrating the importance of additional fertilizer requirements. The findings provide

¹ Of the total costs of production, estimates from Ministry of Agriculture surveys (Burke et al. 2017, ZARI various years) fertilizer costs account for almost 45–55% followed by labor costs involved in land preparation, planting, weeding, post-harvest handling activities; and seed costs around 8–10%; other inputs (plant protection etc.) around 3–5% and transport/marketing etc. accounts for the rest.

compelling evidence for moving beyond blanket fertilizer recommendations toward more nuanced, location-specific approaches. Further GSSAT2 simulations also offered the much-needed temporal and spatial precision to the data i.e., it helped to identify optimal planting windows (November–December) while accounting for regional variations and with the tool's ability to incorporate local soil analysis data (demonstrated with Monze's soil carbon content differences) shows potential for “hyper-localization” of recommendations.

While we can demonstrate the utility of GSSAT2 as a decision tool for spatial nutrient management in this research paper, there are opportunities for further improvement for enhanced calibration and precision. Systematic integration of higher resolution weather data expanded local soil information (from soil testing/analysis data) and additional economic parameters including labor costs besides input costs can progressively improve model's accuracy for hyper-localized recommendations.

Acknowledgements This research was funded by the U.S. Agency for International Development (USAID) Bureau for Resilience, Environment, and Food Security (REFS)-funded Sustainable Opportunities for Improving Livelihoods with Soils (SOILS) Consortium-Space to Place (S2P) initiative, implemented by IFDC.

Author contributions K.S.E., U.S. and L.N. conceptualized the study and methodology; N.B., W.P. and K.S. worked on GSSAT2 Development; U.S. and late P.W. designed the initial GSSAT; K.S.E., J.F.P., L.N. and U.S. wrote the first draft of this manuscript; L.N., N.B., J.F.P., W.P., F.B., K.S., M.D., P.N.M.H., B.K., B.P.M., W.L., Y.G., Z.S., U.S. reviewed and significantly edited and improved the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

- Arouna A, Fatognon IA, Saito K, Futakuchi K (2021) Moving toward rice self-sufficiency in sub-Saharan Africa by 2030: lessons learned from 10 years of the coalition for African Rice Development. *World Development Perspectives* 21:100291. <https://doi.org/10.1016/j.wdp.2021.100291>
- Beck HE, Zimmermann NE, McVicar TR, Vergopolan N, Berg A, Wood EF (2018) Present and future Köppen–geiger climate classification maps at 1-km resolution. *Sci Data* 5(1):180214
- Beyene Y, Semagn K, Crossa J et al (2016) Improving maize grain yield under drought stress and non-stress environments in sub-Saharan Africa using marker-assisted recurrent selection. *Crop Sci* 56(1):344–353
- Bombana NP (2023) Advancing spatial simulations in agriculture: Introducing GSSAT2, an enhanced DSSAT-Based tool. MSc Thesis, University of Passo Fundo, Brazil
- Buresh RJ, Castillo RL, Torre JCD, Laureles EV, Samson MI, Sinohin PJ, Guerra M (2019) Site-specific nutrient management for rice in the Philippines: calculation of field-specific fertilizer requirements by rice crop manager. *Field Crop Res* 239:56–70
- Burke WJ, Jayne TS, Black JR (2017) Factors explaining the low and variable profitability of fertilizer application to maize in Zambia. *Agric Econ* 48:115–126
- Canning D, Sangeeta R, Abdo YS (2015) Africa's demographic transition: Dividend or disaster? World Bank and Agence Française de Développement, Washington
- Chapoto A, Chabala LM, Lungu ON (2016) A long history of low productivity in Zambia: is it time to do away with blanket recommendations? *Zamb Soc Sci J* 6(2):6
- Chikowo DR (2016) Application of the GYGA approach to Zambia. *Global Yield Gap Atlas*. <http://www.yieldgap.org/zambia>. Accessed 4 Sept 2024
- Chivenge P, Zingore S, Ezui KS, Njoroge S, Bunquin MA, Dobermann A, Saito K (2022) Progress in research on site-specific nutrient management for smallholder farmers in sub-Saharan Africa. *Field Crop Res* 281:108503. <https://doi.org/10.1016/j.fcr.2022.108503>
- Corbeels M, Palmer A, Akponikpè PB, Vanlauwe B (2016) Performance and sensitivity of the DSSAT crop growth model in simulating maize yield under conservation agriculture. *Eur J Agron* 76:41–53
- Dobermann A, Witt C, Dawe D et al (2002) Site-specific nutrient management for intensive rice cropping systems in Asia. *Field Crops Res* 74(1):37–66
- Dobermann A, Witt C, Dawe D (2004) Increasing productivity of intensive rice systems through site-specific nutrient management. Science Publishers Inc, IIRI
- Dutta S, Chakraborty S, Banerjee H, Goswami R, Majumdar K, Li B, Jat ML (2020) Maize yield in smallholder agriculture system—an approach integrating socio-economic and crop management factors. *PLoS One* 15(2):e0229100. <https://doi.org/10.1371/journal.pone.0229100>
- FAO (2018) The future of food and agriculture: Alternative pathways to 2050. Summary version. Rome. 64 pp. (also available at <http://www.fao.org/3/CA1553EN/ca1553en.pdf>).
- FAO (2020) Statistical databases. Food and Agriculture Organization of the United Nations, Rome
- FAO, IFAD, UNICEF, WFP, WHO (2022) The state of food security and nutrition in the world 2022: repurposing food and agricultural policies to make healthy diets more affordable. FAO, Rome. <https://doi.org/10.4060/cc0639en>
- Folberth C, Yang H, Gaiser T, Abbaspour KC, Schulin R (2013) Modeling maize yield responses to improvement in nutrient, water and cultivar inputs in sub-Saharan Africa.

- Agric Syst 119:22–34. <https://doi.org/10.1016/j.agry.2013.04.002>
- Gu J, Yang J (2022) Nitrogen (N) transformation in paddy rice field: its effect on N uptake and relation to improved N management. *Crop Environ* 1:7–14. <https://doi.org/10.1016/j.crope.2022.03.003>
- Hagblade S, Diarra A, Jiang W, Assima A, Keita N, Traoré A, Traoré M (2019) Fraudulent pesticides in West Africa: a quality assessment of glyphosate products in Mali. *Int J Pest Manag* 67(1):32–45. <https://doi.org/10.1080/09670874.2019.1668076>
- Hoogenboom G, Porter CH, Boote KJ, Shelia V, Wilkens PW, Singh U, White JW, Asseng S, Lizaso JJ, Moreno LP, Pavan W, Ogoshi R, Hunt LA, Tsuji GY, Jones JW (2019) The DSSAT crop modeling ecosystem. In: Boote KJ (ed) *Advances in crop modeling for a sustainable agriculture*. Burleigh Dodds Science Publishing, Cambridge, pp 173–216. <https://doi.org/10.19103/AS.2019.0061.10>
- Hu C, Sadras VO, Lu G, Zhang P, Han Y, Liu L, Xie J, Yang X, Zhang S (2021) A global meta-analysis of split nitrogen application for improved wheat yield and grain protein content. *Soil Tillage Res* 213:105111
- IFASTAT (2019) Comparative scope of the supply and consumption data sets. <https://www.ifastat.org>
- International Monetary Fund (IMF) (2024) Regional economic outlook for sub-Saharan Africa. <https://www.imf.org/en/Publications/REO/SSA/Issues/2024/04/19/regional-economic-outlook-for-sub-saharan-africa-april-2024?cid=bl-com-SM2024-REOAFREA2024001>
- Janssen BH, Guiking FCT, van der Eijk D, Smaling EM, Wolf J, van Reuler H (1990) A system for quantitative evaluation of the fertility of tropical soils (QUEFTS). *Geoderma* 46(4):299–318
- Johnston AM, Bruulsema TW (2014) 4r nutrient stewardship for improved nutrient use efficiency. *Procedia Eng* 83:365–370
- Jones CA, Kiniry J (1986) CERES-maize: a simulation model of maize growth and development. Texas A & M University Press, College Station
- Jones JW, Hoogenboom G, Porter CH, Boote KJ, Batchelor WD, Hunt LA, Wilkens PW, Singh U, Gijsman AJ, Ritchie JT (2003) The DSSAT cropping system model. *Eur J Agron* 18(3–4):235–265
- Joshi VR, Villalobos C, Castillo Romero O, Porter CH, Hoogenboom G (2021) Pythia: a gridded modeling framework for DSSAT-Cropping System Model for multi-spatial scale applications [Abstract]. ASA, CSSA, SSSA International Annual Meeting, Salt Lake City, UT. <https://scisoc.confex.com/scisoc/2021am/meetingapp.cgi/Paper/134115>
- Keating BA, Carberry PS, Hammer GL, Probert ME, Robertson MJ, Holzworth D, Huth NI, Hargreaves JNG, Meinke H, Hochman Z, McLean G (2003) An overview of APSIM, a model designed for farming systems simulation. *Eur J Agron* 18(3–4):267–288
- Lipper L, Thornton P, Campbell BM, Baedeker T, Braimoh A, Bwalya M, Caron P, Cattaneo A, Garrity D, Henry K, Hottle R, Jackson L, Jarvis A, Kossam F, Mann W, McCarthy N, Meybeck A, Neufeldt H, Remington T, Sen PT, Sessa R, Shula R, Tibu A, Torquebiau EF (2014) Climate-smart agriculture for food security. *Nat Clim Change* 4:1068
- Lusaka Times (2024) President Hichilema declares drought as a national disaster and emergency. Lusaka Times. <https://www.lusakatimes.com/2024/03/01/president-hichilema-declares-drought-as-a-national-disaster-and-emergency/>
- Ma Q, Wang M, Zheng G et al (2021) Twice-split application of controlled-release nitrogen fertilizer met the nitrogen demand of winter wheat. *Field Crops Res* 267:108163
- Ministry of Agriculture (2024) The 2023/2024 agricultural production season crop forecasting survey results report. Government of the Republic of Zambia, Lusaka
- Pavan W, Postal Bombana N, Singh U, Fugice J Jr, Nagaranjan L (2023) Spatial crop modeling at scale: GSSAT2 [Abstract]. In: ASA, CSSA, SSSA international annual meeting, St. Louis, MO. <https://scisoc.confex.com/scisoc/2023am/meetingapp.cgi/Paper/153599>
- Poggio L, De Sousa LM, Kempen B, Heuvelink GBM, Batjes NH (2021) SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil* 7(1):217–240
- Rurinda J, Zingore S, Jibrin JM et al (2020) Science-based decision support for formulating crop fertilizer recommendations in sub-Saharan Africa. *Agric Syst* 180:102790
- Shankar T, Malik GC, Banerjee M, Ghosh A (2014) Nutrient optimization on growth and productivity of rice in the red and lateritic belt of West Bengal. *J Agric Sci Technol* 16(1):12–19
- Timsina J (2018) Can organic sources of nutrients increase crop yields to meet global food demand? *Agron* 8:214. <https://doi.org/10.3390/agronomy8100214>
- Timsina J, Singh VK, Majumdar K (2013) Potassium management in rice-maize systems in South Asia. *J Soil Sci Plant Nutr*. <https://doi.org/10.1002/jpln.201200253>
- Timsina J, Dutta S, Devkota K, Chakraborty S, Neupane RK, Bishta S, Amgain LP, Singh V, Islam S, Majumdar K (2021) Improved nutrient management in cereals using nutrient expert and machine learning tools: productivity, profitability and nutrient use efficiency. *Agric Syst* 192:103181
- United Nations Department of Economic and Social Affairs, Population Division (2019) World population prospects 2019: Highlights (ST/ESA/SER.A/423)
- van Ittersum MK, van Bussel LGJ, Wolf J, Grassini P, van Wart J, Guilpart N, Claessens L, de Groot H, Wiebe K, Mason-D'Croz D, Yang H, Boogaard H, van Oort PAJ, van Loon MP, Saito K, Adimo O, Adjei-Nsiah S, Agali A, Bala A, Chikowo R, Kaizzi K, Kouressy M, Makoi JHJR, Ouattara K, Tesfaye K, Cassman KG (2016) Can sub-Saharan Africa feed itself? *Proc Natl Acad Sci U S A* 113:14964–14969
- Vanlauwe B, Dobermann A (2020) Sustainable intensification of agriculture in sub-Saharan Africa: first things first! *Front Agric Sci Eng* 7:376–382
- Varshini SV, Babu R (2021) Effect of graded levels and split application of nitrogen on plant nutrient uptake, post-harvest soil available nutrients and economics of hybrid maize (Co MH-6). *Environ Ecol* 39(1):92–98

- Waldman KB, Vergopolan N, Attari SZ, Sheffield J, Estes LD, Caylor KK, Evans TP (2019) Cognitive biases about climate variability in smallholder farming systems in Zambia. *Weather Clim Soc* 11(2):369–383. <https://doi.org/10.1175/WCAS-D-18-0050.1>
- World Population Review (2024) World population prospects (2024 revision)—United Nations population estimates and projections: sub-Saharan Africa population 2024. <https://worldpopulationreview.com/continents/sub-saharan-africa>
- Zhao C, Liu B, Piao S et al (2017) Temperature increase reduces global yields of major crops in four independent estimates. *Proc Natl Acad Sci U S A* 114(35):9326–9331. <https://doi.org/10.1073/pnas.1701762114>

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